Chapter 2 — Force and Motion

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. Which is a vector quantity?

(a) time **(b)** momentum **(c)** mass **(d)** energy

2. Which is a scalar quantity?

(a) speed **(b)** displacement **(c)** force **(d)** velocity

3. The x-component of vector A at angle θ is:

(a) A cos θ **(b)** A tan θ **(c)** A/ cos θ **(d)** A sin θ

4. The magnitude of a vector from components is:

(a) $\sqrt{(Ax^2 + Ay^2)}$ **(b)** Ax − Ay **(c)** Ax + Ay **(d)** Ax Ay

5. The dot product A·B equals:

(a) AB sin θ **(b)** AB cos θ **(c)** AB tan θ **(d)** A/B

6. The cross product magnitude |A×B| equals:

(a) AB cos θ **(b)** AB sin θ **(c)** A/B **(d)** AB tan θ

7. The dot product of perpendicular vectors is:

(a) maximum **(b)** negative **(c)** AB **(d)** zero

8. The cross product of parallel vectors is:

(a) AB/2 **(b)** maximum **(c)** zero **(d)** AB

9. The dot product is maximum when vectors are:

(a) perpendicular **(b)** opposite **(c)** at 45° **(d)** parallel

10. The cross product is maximum when vectors are:

(a) perpendicular **(b)** at 30° **(c)** parallel **(d)** opposite

11. Work done is an example of:

(a) cross product **(b)** dot product **(c)** subtraction **(d)** vector addition

12. Torque is an example of:

(a) dot product **(b)** scalar product **(c)** division **(d)** cross product

13. The direction of A×B is given by:

(a) right-hand rule **(b)** Fleming rule **(c)** Lenz rule **(d)** left-hand rule

14. The first equation of motion is:

(a) $v = u + at$ **(b)** $v^2 = u^2 + 2as$ **(c)** $s = ut + \frac{1}{2}at^2$ **(d)** $s = vt$

15. The second equation of motion is:

(a) $s = at$ **(b)** $s = ut + \frac{1}{2}at^2$ **(c)** $v = u + at$ **(d)** $v^2 = u^2 + 2as$

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16. The third equation of motion is:

(a) $s = ut + \frac{1}{2}at^2$ **(b)** $s = \frac{1}{2}at^2$ **(c)** $v = u + at$ **(d)** $v^2 = u^2 + 2as$

17. The slope of a velocity–time graph gives:

(a) force **(b)** displacement **(c)** acceleration **(d)** momentum

18. The area under a velocity–time graph gives:

(a) acceleration **(b)** power **(c)** force **(d)** displacement

19. For a freely falling body, acceleration is:

(a) g **(b)** v/t **(c)** zero **(d)** u/t

20. At the highest point of a projectile, vertical velocity is:

(a) maximum **(b)** g **(c)** zero **(d)** u

21. At the highest point of a projectile, acceleration is:

(a) g upward **(b)** horizontal **(c)** g downward **(d)** zero

22. The horizontal velocity of a projectile:

(a) decreases **(b)** becomes zero **(c)** remains constant **(d)** increases

23. Time of flight of a projectile is:

(a) $2u \sin\theta/g$ **(b)** $2u \cos\theta/g$ **(c)** $u^2 \sin 2\theta/g$ **(d)** $u \sin\theta/g$

24. Maximum height of a projectile is:

(a) $u^2 \sin^2\theta/(2g)$ **(b)** $u^2 \sin 2\theta/g$ **(c)** $u^2 \cos^2\theta/(2g)$ **(d)** $2u \sin\theta/g$

25. Range of a projectile is:

(a) $u \sin\theta/g$ **(b)** $2u \sin\theta/g$ **(c)** $u^2 \sin 2\theta/g$ **(d)** $u^2 \sin^2\theta/(2g)$

26. Range is maximum at angle:

(a) 90° **(b)** 45° **(c)** 60° **(d)** 30°

27. Two angles giving the same range are:

(a) complementary **(b)** supplementary **(c)** perpendicular **(d)** equal

28. The path of a projectile without air resistance is:

(a) circle **(b)** parabola **(c)** straight line **(d)** ellipse

29. Momentum is:

(a) mv **(b)** $\frac{1}{2}mv^2$ **(c)** mgh **(d)** ma

30. Impulse equals:

(a) mass × acceleration **(b)** change in energy **(c)** force × distance **(d)** change in momentum

31. SI unit of impulse is:

(a) W s **(b)** N s **(c)** N m **(d)** J s

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32. Rate of change of momentum is:

(a) impulse **(b)** power **(c)** energy **(d)** force

33. In an isolated system, total momentum is:

(a) zero **(b)** conserved **(c)** increased **(d)** decreased

34. In an elastic collision:

(a) kinetic energy is lost **(b)** bodies stick **(c)** kinetic energy is conserved **(d)** momentum is lost

35. In a perfectly inelastic collision, bodies:

(a) stick together **(b)** exchange velocities **(c)** stop always **(d)** separate with same speed

36. Airbags reduce injury by:

(a) removing impulse **(b)** decreasing impact time **(c)** increasing momentum change **(d)** increasing impact time

37. A karate expert breaks bricks by:

(a) short impact time **(b)** long impact time **(c)** zero force **(d)** small momentum

38. Rocket propulsion is based on:

(a) Bernoulli principle **(b)** Archimedes principle **(c)** conservation of energy only **(d)** conservation of momentum

39. A rocket can move in vacuum because:

(a) it pushes air **(b)** it uses friction **(c)** it has wings **(d)** it pushes its own gases

40. As fuel burns, rocket acceleration:

(a) increases **(b)** becomes zero **(c)** remains same **(d)** decreases

41. If $|A+B| = |A-B|$, the angle between A and B is:

(a) 0° **(b)** 90° **(c)** 180° **(d)** 45°

42. The scalar product of two vectors is maximum if they are:

(a) parallel **(b)** perpendicular **(c)** at 60° **(d)** at 30°

43. The angle at which dot product equals cross product magnitude is:

(a) 90° **(b)** 45° **(c)** 60° **(d)** 30°

44. The acceleration at the top of a projectile trajectory is:

(a) g **(b)** zero **(c)** maximum **(d)** minimum

45. The range of a projectile is same for angles that are:

(a) perpendicular **(b)** complementary **(c)** equal **(d)** supplementary

46. A body at rest may have:

(a) momentum **(b)** energy **(c)** speed **(d)** velocity

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47. When a ball is thrown upward, the conservative force is:

(a) hand contact **(b)** air resistance **(c)** friction **(d)** gravity

48. The SI unit of momentum is:

(a) kg m s^{-1} **(b)** J s **(c)** N m **(d)** W s

49. The dimensions of impulse are:

(a) $[MLT^{-2}]$ **(b)** $[ML^2T^{-2}]$ **(c)** $[MLT^{-1}]$ **(d)** $[ML^2T^{-3}]$

50. If momentum is doubled, kinetic energy becomes:

(a) four times **(b)** unchanged **(c)** two times **(d)** half

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Answer Key with Brief Reasons

1. Answer: (b) momentum — Momentum has magnitude and direction.

2. Answer: (a) speed — Speed has magnitude only.

3. Answer: (a) A cos θ — Adjacent component uses cosine.

4. Answer: (a) √(Ax² + Ay²) — Pythagoras gives magnitude.

5. Answer: (b) AB cos θ — Dot product uses cosine.

6. Answer: (b) AB sin θ — Cross product uses sine.

7. Answer: (d) zero — cos 90° = 0.

8. Answer: (c) zero — sin 0° = 0.

9. Answer: (d) parallel — cos 0° = 1.

10. Answer: (a) perpendicular — sin 90° = 1.

11. Answer: (b) dot product — W = F·d.

12. Answer: (d) cross product — τ = r×F.

13. Answer: (a) right-hand rule — Right-hand rule gives normal direction.

14. Answer: (a) v = u + at — Uniform acceleration gives v = u + at.

15. Answer: (b) s = ut + ½at² — Area under v–t graph gives s = ut + ½at².

16. Answer: (d) v² = u² + 2as — Eliminating t gives $v^2 = u^2 + 2as$.

17. Answer: (c) acceleration — Slope = Δv/Δt.

18. Answer: (d) displacement — Area = velocity × time.

19. Answer: (a) g — Gravity gives constant g near Earth.

20. Answer: (c) zero — Vertical motion momentarily stops.

21. Answer: (c) g downward — Gravity still acts.

22. Answer: (c) remains constant — No horizontal acceleration without air resistance.

23. Answer: (a) 2u sin θ/g — T = 2u sin θ/g.

24. Answer: (a) u² sin²θ/(2g) — $H = u^2 \sin^2\theta/(2g)$.

25. Answer: (c) u² sin 2θ/g — $R = u^2 \sin 2\theta/g$.

26. Answer: (b) 45° — sin 2θ is maximum at 45°.

27. Answer: (a) complementary — θ and 90° − θ give same sin 2θ.

28. Answer: (b) parabola — Constant horizontal velocity and vertical acceleration give parabola.

29. Answer: (a) mv — p = mv.

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30. Answer: (d) change in momentum — $F\Delta t = \Delta p$.

31. Answer: (b) N s — Impulse = force × time.

32. Answer: (d) force — Newton's second law: $F = \Delta p/\Delta t$.

33. Answer: (b) conserved — No external force means momentum conservation.

34. Answer: (c) kinetic energy is conserved — Elastic collisions conserve KE and momentum.

35. Answer: (a) stick together — They move with common velocity.

36. Answer: (d) increasing impact time — Larger Δt reduces average force.

37. Answer: (a) short impact time — Short Δt gives large force.

38. Answer: (d) conservation of momentum — Ejected gases give forward momentum.

39. Answer: (d) it pushes its own gases — Action–reaction is between rocket and gases.

40. Answer: (a) increases — Mass decreases for same thrust.

41. Answer: (b) 90° — Equality implies dot product zero.

42. Answer: (a) parallel — $\cos 0° = 1$.

43. Answer: (b) 45° — $\cos\theta = \sin\theta$ at 45°.

44. Answer: (a) g — Gravity acts throughout.

45. Answer: (b) complementary — θ and $90° - \theta$.

46. Answer: (b) energy — It can have potential or internal energy.

47. Answer: (d) gravity — Gravity is conservative.

48. Answer: (a) kg m s^{-1} — $p = mv$.

49. Answer: (c) [MLT^{-1}] — Impulse = momentum.

50. Answer: (a) four times — $K = p^2/2m$.

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